

# ARGUMENTATION LAYER ANNOTATION GUIDELINES

*(Adapted from Stab & Gurevych (2015), extended with implicit argumentation)*

## 1. Introduction

The present guidelines describe the annotation of argumentative components in two types of texts: narrative texts (STORAL folktales) and spoken discourse (TED talk transcripts). The objective is to identify and annotate the fundamental segments of arguments, namely major claims, claims, and premises, while extending existing frameworks to include the distinction between explicit and implicit argumentation.

This work is based on the framework proposed by Stab and Gurevych (2015), which provides a well-established model for annotating argument components in persuasive essays. However, the present annotation scheme introduces two key modifications. First, it omits the annotation of argumentative relations (e.g., support and attack), focusing exclusively on the identification of argument components. Second, it introduces an additional layer distinguishing between explicitly stated and implicitly conveyed argumentative content.

This results in a simplified yet conceptually enriched annotation scheme that is particularly suited for narrative texts, where argumentation is often not overtly expressed.

*Some things to note before starting:*

- The scheme allows partial component structures
- Not every text must contain all three component types
- In short narrative texts, especially STORAL, the structure may collapse into a major claim alone or a major claim supported by a premise-like rationale
- The absence of a separate claim is an expected and valid outcome, not an annotation failure

## 2. Overview of the Annotation Scheme

### 2.1 Argument Components

In line with classical argumentation theory, an argument can be understood as a structure consisting of a claim supported by one or more premises (Stab & Gurevych, 2014). The claim represents a proposition that is subject to debate, while the premises provide justification for accepting or rejecting this proposition.

Within this framework, three types of argument components are distinguished:

- Major Claim, representing the overall standpoint or main message of a text
- Claim, representing intermediate statements that support or develop the major claim
- Premise, representing reasons, evidence, or justifications that underpin claims

These components are annotated independently, without explicitly encoding the relations between them.

## 2.2 Explicit and Implicit Argumentation

A central extension of this annotation scheme is the distinction between explicit and implicit argumentation. Implicit argumentation arises when the logical connection between statements depends on shared background knowledge or contextual inference. As discussed in prior work, many arguments rely on unstated premises or conclusions that are nonetheless crucial for understanding the reasoning process (Singh et al., 2021).

To capture this phenomenon, each argument component is annotated as either:

- Explicit, if it is directly stated in the text
- Implicit, if it must be inferred from context

*Additional rules:*

- An implicit component should be annotated only when it is necessary to make sense of the text's argumentative meaning, and when it is strongly suggested by the text as a whole, not merely possible as one of several possible interpretations
- Only annotate one best-supported implicit formulation per component role
- Do not generate multiple alternative implicit readings during annotation
- Do not annotate an implicit claim or premise if it simply restates the same idea as the major claim in different words
- If the text only supports a broad takeaway, annotate only the major claim
- If the inferred premise is only generic world knowledge with no strong textual grounding, do not annotate it

### 2.2.1 Annotating Explicit Components

Explicit argument components are directly stated in the text:

***Procedure:***

- Select the minimal span of text that expresses a complete argumentative unit
- The span should correspond to either:
  - A full sentence
  - A clause containing a complete statement
- Avoid selecting:
  - Unnecessary surrounding text
  - Discourse markers (e.g., "I think that...") unless they affect the meaning

- Then assign one of the respective labels (MC-EXP/ C-EXP/ P-EXP)

## 2.2.2 Annotating Implicit Components

Implicit argument components are not directly stated in the text, but they are necessary to understand its argument meaning.

Because there is no direct span that expresses them, annotation requires both:

- Selecting a supporting span
- Writing an explicit formulation/interpretation

### *Step 1: select a supporting span*

Annotators must select a span of text that best supports the inference. The span should:

- Be the part of the text from which the implicit meaning is derived
- Reflect the key event or statement that leads to the inference

It should not:

- Try to “cover” the whole story unnecessarily
- Be unrelated to the inferred meaning

### *Step 2: assign labels*

Assign one of the respective labels (MC-IMP/ C-IMP/ P-IMP)

### *Step 3: write explicit formulation*

For every implicit annotation, annotators must write a clear sentence that makes the component explicit.

The sentence should:

- Express the inferred meaning as a complete statement
- Be generalizable
- Stay as close as possible to what is supported by the text
- Avoid adding unnecessary interpretation

## 2.3 Annotation Labels

The combination of component type and realization results in six annotation labels:

- Explicit Major Claim (MC-EXP)
- Implicit Major Claim (MC-IMP)
- Explicit Claim (C-EXP)
- Implicit Claim (C-IMP)
- Explicit Premise (P-EXP)
- Implicit Premise (P-IMP)

## 2.4 Scope of Annotation

The annotation process is limited to the identification of argument components. In contrast to the original framework by Stab and Gurevych, this scheme does not include the annotation of argumentative relations such as support or attack. The focus is therefore on what is being argued, rather than how components are connected.

## 2.5 Annotation Procedure in INCEpTION

Annotation in this project is carried out using the INCEpTION annotation tool. The process consists of steps that must be followed consistently for all texts.

Each annotation involves:

1. Selecting a text span
2. Assigning a label to the span
3. (For implicit components only) Writing an explicit formulation of the component

# 3. Annotation of Argument Components

## 3.1 General Principles

The annotation of argument components is conducted at the level of textual spans. An argument component may correspond to a full sentence or to a smaller clause within a sentence. It is important to note that sentences may include more than one argument component, which should be annotated separately when they represent distinct propositions.

Following the principles outlined by Stab and Gurevych, annotated spans should represent complete and meaningful statements. Expressions that do not contribute to the propositional content of an argument, such as discourse markers or meta-commentary (e.g., “I believe that”, “it is clear that”), should generally be excluded from the annotation unless they alter the meaning of the statement (e.g., through negation).

**Punctuation Rule:** Punctuations at the end of an argument component should not be included in the annotation.

## 3.2 Annotation of Major Claims

The major claim represents the central message or overarching conclusion of a text. In persuasive essays the major claim represents the stance of the author about the essay topic. It is also called a thesis statement and frequently indicated by opinion expressions like “From my point of view...”, “In my opinion...”, “I strongly believe that...”, etc. Usually, the major claim is present in the introduction or conclusion of an essay or in both. In narrative texts such as folktales, the major claim often takes the form of a moral or lesson derived from the events of the story.

Major claims may be explicitly stated or implicitly conveyed. Explicit major claims are typically identifiable through declarative statements that summarize the main point of the text. Implicit major claims, on the other hand, must be inferred from the narrative or argumentative structure.

Consider the following example from a TED talk:

**Example 3.2.1 TED\_001:** “[...] we strongly believe that [by reimagining the role and the agency of the architect, we have set up the core idea of the project into motion; that is, to transform the river from sewage to public space for all, thereby making sure that the city of Fez will remain a living city for its inhabitants rather than a mummified heritage]\_[MC-EXP].”

This statement clearly expresses the speaker’s central takeaway in the conclusion section, and is therefore annotated as an **explicit major claim (MC-EXP)**.

In contrast, consider the following example from a folktale:

**Example 3.2.2 STORAL\_001:** “A man [...] begged all the trees to give him a small branch which he wanted for a particular purpose. The trees were good-natured and [gave him one of their branches]\_[MC-IMP]. What did the man do but fix it into the axe head, and soon set to work [cutting down tree after tree]\_[MC-IMP].”

In this case, the text does not explicitly state a general conclusion. However, the events strongly suggest a broader lesson, such as: *“Helping an enemy can lead to one’s own destruction”*

**Example 3.2.3. STORAL\_020:** “God’s purpose are always just. you thought that devoting yourself to god means you can judge other’s. because of this while you filled your heart with impurity of other’s sin, [that lady shed away her sin with pure repentance and her soul was so light that she went to heaven and you soul weighed down with stones is too heavy to lift]\_[MC-IMP].”

In this example, it can be inferred from the ending (God’s judgment): *“Being truly sorry for your actions matters more than appearing religious or judging others”*.

**Example 3.2.4. STORAL\_019:** “[...]“My friends, when I finish eating, I am accustomed to singing.. now I shall sing..” Camel got scared and said, “Don’t sing yet. I haven’t finished eating vegetables and if you sing now farmer will hear you and we will be in trouble.”[Fox didn’t listen to him and started singing]\_[MC-IMP], hearing it farmer came out running with a very large stick. Fox was small and quickly ran away [but camel was slow and received many blows on his legs and back before being able to escape from farmer]\_[MC-IMP]. [...] “[When I am done eating, I am accustomed to take bath]\_[MC-IMP].” Listening to this fox got scared and they were in middle of river where water was deep and fox could have easily drowned. He pleaded, “Please.. don’t take bath here. if you take bath I will be drowned in river.”[Camel ignored him and

lowered his back deep into river until the fox lost her grip on his back and began to splash around helplessly against the fast current]\_[MC-IMP]. “Help me..!! help me..!!”, cried fox. Camel got up and said, “[Now you understand?? are you sorry that because of your selfishness I got beaten by farmer.” Fox cried, “Yes.. yes.. I have realized my mistake.. now please forgive me]\_[MC-IMP].” Seeing this camel felt pity and swam the rest of the way across the river and climbed up the bank onto the warm grass.”

In this example, the main claim is implied, and can be derived from the full narrative and the actions taken. The fox acts selfishly which results in harming the camel. The camel then retaliates and forces the fox to realize its mistake: “*Selfish behavior can harm others.*”

Since this conclusion is not directly stated but can be inferred from the narrative, it is annotated as an **implicit major claim (MC-IMP)**.

### 3.3 Annotation of Claims

Claims represent intermediate statements that contribute to the development of the major claim. They are typically evaluative or interpretive in nature and express a viewpoint that requires justification.

Explicit claims are directly stated in the text and can often be identified through evaluative language or discourse markers. For example:

**Example 3.3.1. TED\_001:** “The entire city is a UNESCO World Heritage Site. Since the 1950s, as the population of the medina grew, **[basic urban infrastructure such as green open spaces and sewage quickly changed and got highly stressed]**. **[One of the biggest casualties of the situation was the Fez River, which bisects the medina in its middle and has been considered for many centuries as the city's very soul]** \_[C-EXP].”

**Example 3.3.2. TED\_001:** “[**This plaza used to be a chaotic transportation hub that actually compromised the urban integrity of the medina]** \_[C-EXP], that has the largest pedestrian network in the world. And right beyond the historic bridge that you can see here, right next to the plaza, you can see that **[the river looked like a river of trash]** \_[C-EXP].”

These statements provide an interpretation of a situation and are therefore annotated as an explicit claim (C-EXP).

Implicit claims, by contrast, are not directly stated but can be inferred from the surrounding context. In the case of finding an implicit claim, it would be annotated as (C-IMP).

*Note:* In STORAL texts, the argumentative structure may be compressed. Because the stories are short and heavily implicit, the annotator may identify:

- Only a major claim
- Or a major claim plus premise

- Without identifying a separate claim because it can happen when the inferred “claim” would simply duplicate the major claim and would not add a distinct intermediate argumentative step

### 3.4 Annotation of Premises

Premises provide the reasoning or evidence that supports a claim. They play a crucial role in establishing the plausibility of an argument.

Explicit premises are directly stated and often introduced by discourse markers such as “because”, “since”, or “for example”. Consider the following example:

**Example 3.4.1. TED\_001:** “[**One of the biggest casualties of the situation was the Fez River**]<sub>[MC-EXP]</sub>, which bisects the medina in its middle and has been considered for many centuries as the city's very soul. In fact, one can witness the presence of the river's extensive water network all throughout the city, in places such as private and public fountains. Unfortunately, [because of the pollution of the river, it has been covered little by little by concrete slabs since 1952]<sub>[P-EXP]</sub>.”

**Example 3.4.2. TED\_001:** Actually, [**the state of the river before entering the medina is pretty healthy**]<sub>[MC-EXP]</sub>. [Then pollution takes its toll, mainly due to untreated sewage and chemical dumping from crafts such as tanning]<sub>[P-EXP]</sub>.”

\*For easier visibility in this documentation:

- **The main/claim is in bold**
- The premise is underlined, to show that it is backing up the claim

These statements clearly provide a causal explanation and are therefore annotated as explicit premises (P-EXP).

Implicit premises, on the other hand, are not explicitly stated but are necessary to complete the reasoning. For example:

**Example 3.4.3. STORAL\_019:** “When I am done eating, I am accustomed to take bath.” Listening to this fox got scared and they were in middle of river where water was deep and fox could have easily drowned. He pleaded, “Please.. don't take bath here. if you take bath I will be drowned in river.” Camel ignored him and lowered his back deep into river until the fox lost her grip on his back and began to splash around helplessly against the fast current]. “Help me..!! help me..!!”, cried fox. Camel got up and said, “[Now you understand?? are you sorry that because of your selfishness I got beaten by farmer.” Fox cried, “Yes.. yes.. I have realized my mistake.. now please forgive me]<sub>[P-EXP]</sub>” Seeing this camel felt pity and swam the rest of the way across the river and climbed up the bank onto the warm grass.”

We can derive that the camel imitates the fox’s behavior: “*Experiencing consequences helps individuals understand what they did wrong.*”

Since this reasoning step is not explicitly stated, it is annotated as an implicit premise (P-IMP).

Implicit premises can be quite common in both narrative and argumentative texts, as they rely on shared knowledge and assumptions between the author and the reader.

## 4. Dataset-Specific Considerations

The two datasets used in this study exhibit different characteristics with respect to argumentation.

The STORAL dataset consists of folktales, which are primarily narrative in nature. As such, argumentation is often implicit, and explicit argumentative markers are rare. Annotators should therefore expect a higher proportion of implicit components and should rely on careful interpretation of narrative events.

In contrast, TED talk transcripts often contain more explicit forms of argumentation, including clearly stated claims and premises. However, implicit reasoning still plays an important role, particularly in persuasive or reflective segments.

## 5. Annotation Procedure

The annotation process consists of the following steps.

1. Read the full text
2. Decide whether the text contains a main takeaway that functions as a major claim
3. Check whether there is a distinct intermediate claim separate from the major claim
4. Check whether there is evidence or reasoning that functions as a premise
5. For each component identified, decide whether it is explicit or implicit
6. If a candidate claim duplicates the major claim, do not annotate it separately

This process requires both linguistic analysis and interpretive judgment. Annotators should aim for consistency and avoid over-interpreting ambiguous cases.

## References

- Singh, K., Inoue, N., Mim, F. S., Naitoh, S., & Inui, K. (2021, October 26). *Annotating Implicit Reasoning in Arguments with Causal Links*. arXiv.org. <https://arxiv.org/abs/2110.13692>



Stab, C., & Gurevych, I. (2014, August). Annotating argument components and relations in persuasive essays. In *Proceedings of COLING 2014, the 25th international conference on computational linguistics: Technical papers* (pp. 1501-1510).

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## Appendix A

### Claim Indicators

accordingly  
as a result  
consequently  
conclude that  
clearly  
demonstrates that  
entails  
follows that  
hence  
implies  
in short  
in conclusion  
indicates that  
it follows that  
it is highly probable that  
it should be clear that  
it should be clear  
points to the conclusions  
proves that  
shows that  
so  
suggests that  
the point I'm trying to make  
therefore  
thus

to sum up  
we may deduce

## Appendix B

### Premise Indicators

assuming that  
as  
as indicated by  
as shown  
besides  
because  
deduced  
derived from  
due to  
for  
for example  
for instance  
for the reason that  
furthermore  
given that  
in addition  
in light of  
in that  
in view of  
indicated by  
is supported by  
may be inferred  
moreover  
researchers found that  
this can be seen from  
since  
since the evidence is  
what's more  
whereas